

Indexed Bellman Information Complexity

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

Abstract

We develop *indexed Bellman information complexity*, a representation-level theory of interactive decision making centered on information indices and reference histories. The representation strips away problem-specific syntax and retains only the ingredients needed for dynamic programming and information accounting, thereby unifying the earlier framework of indexed algorithmic information ratios (AIR). On the upper-bound side, regret is controlled by Bellman supersolutions or potential identities whose gradient bracket is paid for by indexed information. Upper-confidence-bound (UCB), estimation-to-decision/decision-estimation-coefficient (E2D/DEC), and adaptive-minimax-sampling or exploration-by-optimization (AMS/EBO) methods appear as three relaxations of this same identity. On the lower-bound side, the posterior-reference trajectory supplies both the information telescope and the ghost quantile of small-regret trajectories. The resulting critical radius in the lower bound is an effective-dimension-scale quantity, as in Fano and local-prior-mass lower bounds, rather than the constant radius of a two-point Le Cam argument. The examples show that DEC is best viewed as a greedy relaxation of indexed Bellman information complexity, not as a universally tight conversion mechanism. We illustrate the framework through several applications, with particular emphasis on kernel bandits. In this setting, the active action marginal provides a concrete basis for comparing UCB, E2D, and AMS/EBO.

Contents

1	Introduction	2
2	Indexed Bellman experiments and the dynamic foundation	4
2.1	Dynamic programming before representation	4
2.2	Latent environments, indices, and losses	5
2.3	Bellman-sufficient representation	5
3	Indexed information and exact AIR/MAIR identities	6
3.1	Posterior-reference histories and index information	6
3.2	Fixed-truth log gain and deterministic specialization	7
3.3	Exact AIR and MAIR regret identities	8
3.4	Upper bounds: one identity and three algorithm families	12
4	Lower bounds: reference histories and quantile indices	14
4.1	Ghost probability and true good probability	14
4.2	Reference-history quantile theorem	15
4.3	Bellman-Fano lower certificates	16
4.4	DEC as one-step greedy relaxation	17

5	Applications and examples	19
5.1	Finite-action kernel bandits: GP-UCB, GP-E2D, and AMS/EBO	19
5.2	Multi-armed bandits lower bound via reference history	24
5.3	Linear bandits lower bound via reference history	25
5.4	Bellman-rank as indexed representation	28
6	Conclusion	29

1 Introduction

Information-theoretic lower bounds compare two quantities. The first is the information revealed by an experiment. The second is the mass, entropy, or packing size of the set on which a learner can be successful. In noninteractive statistical estimation this comparison is especially transparent: the experiment is fixed in advance, and local-prior-mass or metric-entropy calculations can be balanced against Kullback-Leibler (KL) information; see, e.g., the information-theoretic upper and lower bounds of Zhang (2006); Yu (1997). In interactive decision making the learner designs the experiment while learning. The action distribution is itself a random consequence of the unknown environment through the history. This adaptivity is exactly what makes a sharp theory difficult.

A powerful response is algorithmic Fano’s method, meaning a Fano-type inequality in which the comparison data set is generated from an algorithm-dependent reference or “ghost” experiment rather than from a fixed nonadaptive sampling scheme. This perspective appears in recent interactive Fano frameworks (Chen et al., 2024) and algorithmic lower bounds (Xu, 2026); it is also closely related to the decision-estimation coefficient (DEC) approach of Foster et al. (2021) and its constrained/localized refinements (Foster et al., 2023). The central obstacle in the algorithmic Fano method is what comes next: collapsing a T -round adaptive experiment into a single one-round action distribution generally requires a localized change-of-measure argument, because the action law under a hard environment need not coincide with the action law under the reference experiment. Such a reduction can obscure or discard the geometry of the posterior trajectory, making the theory more rigid than necessary.

This paper leverages indexed posterior-reference histories to construct a lower-bound counterpart of the information telescopes appearing in the upper-bound theory. Given a prior μ , the Bayesian mixture law generates a reference history H'_T and a posterior process. The indexed information gain is evaluated on that same reference history. If $Y = \chi(\Omega)$ is the chosen information index, then the exact identity is

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t),$$

where S'_t is the reference posterior state and p'_t is the algorithm’s action distribution at that reference history. The same reference history defines the ghost probability that the algorithm has small target-wise average regret. Thus, the lower-bound theorem takes a stepwise form that mirrors the algorithmic information ratio (AIR) and model-index algorithmic information ratio (MAIR) telescopes in our upper-bound theorems.

$$\text{if } T\bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad \text{then } \mathbb{E}_\mu^{\text{Alg}} \text{Reg}_T \geq \frac{Tr}{2}.$$

Here r is an average-regret threshold, p_r^{Alg} is the posterior-reference ghost probability of average regret at most r , and $\bar{C}_\chi^{\text{Alg}} = T^{-1}\mathbb{E} \sum_t \mathcal{I}_\chi(S'_t, p'_t)$ is average indexed information. The average

notation changes nothing mathematically; it displays the horizon explicitly. The critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r}.$$

When the ghost-good mass is roughly $\exp(-d_{\text{eff}})$, the admissible total information is d_{eff} -level. This is the Fano/local-entropy radius, not the constant two-point radius. In other words, we identify the posterior-reference indexed-information coordinates in which interactive Fano and AIR/MAIR upper bounds become two sides of the same Bellman information complexity.

The word “Bellman” in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman’s principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is a Bellman subsolution or, more flexibly, a Bellman-Fano certificate that combines an information supersolution with a ghost-good-mass supersolution. One-round DEC/E2D is recovered only after replacing the dynamic continuation value by a greedy offset. It may be sharp in simple geometries, but it is not the minimal object for matching upper and lower bounds.

Main contributions.

1. We formulate *indexed Bellman experiments*, a representation-level framework in which the state is sufficient for losses, observation kernels, posterior predictive laws, posterior updates, and indexed information increments. This strictly generalizes DMSO (Foster et al., 2021) and explores minimal structural specification for sequential learnability.
2. We prove the indexed posterior-reference chain rule and the AIR/MAIR regret identities. AIR corresponds to the decision index $Y = \pi^*(\Omega)$; MAIR corresponds to the model or environment index $Y = \Omega$. We record simplifications and specializations, the gradient brackets, and the reason Danskin’s theorem appears in the original AIR proof of Xu and Zeevi (2025).
3. We give a unified “one identity, three algorithms” theorem. UCB methods (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010) certify the bracket through self-normalized calibration and optimism. E2D/DEC methods (Foster et al., 2021, 2023) optimize a one-step offset. AMS/EBO methods (Lattimore and Györfy, 2021; Xu and Zeevi, 2025) control a robust saddle version of the same bracket.
4. We prove the reference-history quantile lower theorem and Bellman lower certificates. As an indexed posterior-reference specialization of the interactive Fano method, the lower side uses the same stepwise indexed information as the upper side, but compares it to ghost-good entropy rather than to a one-round coefficient. Through bandit examples we show DEC/E2D is best viewed as a greedy relaxation of indexed Bellman information complexity, not as a universally tight conversion coefficient.
5. We develop several representative examples, including kernel bandits with an indexable action set, multi-armed bandits, linear bandits, and a Bellman-rank embedding. The kernel-bandit example highlights a regime where the optimal active action is finite, even though the unknown RKHS function remains infinite-dimensional. This makes indexed AIR or optimism the natural theory, while global DEC may be vacuous unless it is properly localized.

Organization. Section 2 defines the dynamic foundation and indexed Bellman experiments. Section 3 gives the indexed information identities and upper algorithmic relaxations. Section 4 gives the quantile lower theorem, Bellman-Fano certificates, and the status of one-round DEC. Section 5 gives applications to kernel bandits and other problems.

2 Indexed Bellman experiments and the dynamic foundation

2.1 Dynamic programming before representation

The minimal representation becomes clear if we begin with the dynamic program. Fix a prior μ over latent environments Ω . At time t , the posterior $b_t = \mathcal{L}_\mu(\Omega \mid H_{t-1})$ summarizes what the Bayesian learner knows. If the history can be compressed into a state s , and if action a produces observation law $P_{s,a}$ and posterior state $\tau(s, a, o)$, then the exact Bayes regret value satisfies

$$V_{T+1}(s) = 0, \quad V_t(s) = \inf_{p \in \Delta(\mathcal{A}_t)} \left\{ \ell(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s,a}} V_{t+1}(\tau(s, a, o)) \right\}. \quad (1)$$

Here

$$\ell(s, p) = \mathbb{E}_{a \sim p} \int \ell_\omega(s, a) b_s(d\omega)$$

is the posterior expected one-step loss. A minimizer in (1) is the Bayes-optimal algorithm for the prior. For minimax regret, Yao's principle (Yao, 1977) gives

$$\mathfrak{R}_T^* := \inf_{\text{Alg}} \sup_{\omega \in \Omega} \mathbb{E}_\omega^{\text{Alg}} L_\omega(H_T) \geq \sup_{\mu} \inf_{\text{Alg}} \mathbb{E}_{\omega \sim \mu} \mathbb{E}_\omega^{\text{Alg}} L_\omega(H_T).$$

Thus Bayes values are lower-bound certificates for minimax values; conversely, uniform Bellman upper relaxations yield minimax algorithms.

Definition 2.1 (Bellman supersolution and subsolution). *Let T_t denote the Bellman operator in (1). A sequence $\bar{V}_t$ is a regret supersolution if*

$$\bar{V}_{T+1} \geq 0, \quad \bar{V}_t(s) \geq (\mathsf{T}_t \bar{V}_{t+1})(s) \quad \text{for all } t, s.$$

A sequence $\underline{V}_t$ is a regret subsolution if

$$\underline{V}_{T+1} \leq 0, \quad \underline{V}_t(s) \leq (\mathsf{T}_t \underline{V}_{t+1})(s) \quad \text{for all } t, s.$$

Proposition 2.2 (Bellman sandwich). *If $\bar{V}$ is a supersolution, then a policy that approximately attains the right-hand side of the supersolution inequality has Bayes regret at most $\bar{V}_1(s_1)$, up to the accumulated approximation errors. If $\underline{V}$ is a subsolution, then every algorithm has Bayes regret at least $\underline{V}_1(s_1)$. Consequently,*

$$\underline{V}_1(s_1) \leq V_1(s_1) \leq \bar{V}_1(s_1).$$

Proof. The upper statement follows by induction. If the algorithm chooses p_t so that

$$\ell(s, p_t) + \mathbb{E} \bar{V}_{t+1}(S_{t+1}) \leq \bar{V}_t(s) + \varepsilon_t,$$

then summing the resulting inequality along the trajectory telescopes the potential and yields regret at most $\bar{V}_1(s_1) + \sum_t \varepsilon_t$. The lower statement is the same induction with inequalities reversed: for any decision kernel p_t ,

$$\ell(s, p_t) + \mathbb{E} \underline{V}_{t+1}(S_{t+1}) \geq \underline{V}_t(s),$$

and summing gives expected loss at least $\underline{V}_1(s_1) - \mathbb{E} \underline{V}_{T+1}(S_{T+1}) \geq \underline{V}_1(s_1)$. $\square$

This proposition is the organizing principle of the paper. The upper theory constructs supersolutions by paying the Bellman bracket with information. The lower theory constructs subsolutions indirectly, by showing that any algorithm with too little information has too small a ghost-good probability to achieve low regret. The representation required by both sides is exactly the representation in which these Bellman backups are well-defined.

2.2 Latent environments, indices, and losses

Let Ω be a latent environment space. At round t , after history

$$H_{t-1} = (A_1, O_1, \dots, A_{t-1}, O_{t-1}),$$

an algorithm chooses $A_t \sim p_t(\cdot \mid H_{t-1}) \in \Delta(\mathcal{A}_t)$ and receives

$$O_t \sim P_{\omega,t}(\cdot \mid H_{t-1}, A_t).$$

The one-step loss is $\ell_t(\omega, H_{t-1}, A_t) \in [0, 1]$. For a fixed algorithm and environment ω , $\mathbb{P}_\omega^{\text{Alg}}$ denotes the trajectory law, and

$$L_\omega(H_T) = \sum_{t=1}^T \ell_t(\omega, H_{t-1}, A_t), \quad \bar{L}_\omega(H_T) = \frac{1}{T} L_\omega(H_T).$$

The information target need not be the full environment. We choose a measurable index

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The model-index choice $Y = \Omega$ gives MAIR. The action-index choice $Y = \pi^*(\Omega)$ gives AIR, where $\pi^*(\Omega)$ denotes an optimal decision under the latent environment. The loss may depend on more than Y , so for index-based analysis we use the conditional loss

$$\ell_t^\chi(y, H_{t-1}, a) := \mathbb{E}_\mu[\ell_t(\Omega, H_{t-1}, a) \mid \chi(\Omega) = y, H_{t-1}], \quad (2)$$

whenever the regular conditional distribution exists. Define

$$L_\chi(y, H_T) = \sum_{t=1}^T \ell_t^\chi(y, H_{t-1}, A_t), \quad \bar{L}_\chi(y, H_T) = \frac{1}{T} L_\chi(y, H_T).$$

Under the true Bayesian experiment,

$$\mathbb{E}_{\Omega \sim \mu, H_T \sim \mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y, H_T} L_\chi(Y, H_T). \quad (3)$$

If one wants a pathwise model-wise event instead of an index-conditioned Bayes event, take $Y = \Omega$.

2.3 Bellman-sufficient representation

Definition 2.3 (Indexed Bellman-sufficient representation). *Fix a prior μ and index map χ . Let $b_t = \mathcal{L}_\mu(\Omega \mid H_{t-1})$ be the posterior under the Bayesian mixture law. A representation is a process*

$$S_t = \phi_t(H_{t-1}, b_t).$$

It is exact Bellman-sufficient if, for every state s , action a , and environment ω , there are objects

$$P_{\omega,s,a} \in \Delta(\mathcal{O}), \quad \ell_\omega(s, a), \quad \tau(s, a, o),$$

such that the following hold.

- (i) Predictive sufficiency: $P_{\omega,t}(\cdot \mid H_{t-1}, a) = P_{\omega,S_t,a}$.
- (ii) Loss sufficiency: $\ell_t(\omega, H_{t-1}, a) = \ell_\omega(S_t, a)$.
- (iii) Posterior predictive sufficiency: $P_{s,a} = \int P_{\omega,s,a} b_s(d\omega)$ is determined by s .
- (iv) Update sufficiency: under the posterior-reference law, $S_{t+1} = \tau(S_t, A_t, O_t)$.
- (v) Index sufficiency: for $q_s = \mathcal{L}(\chi(\Omega) \mid S_t = s)$ and $b_s^y = \mathcal{L}(\Omega \mid S_t = s, \chi(\Omega) = y)$, the conditional predictive laws

$$P_{s,a}^y := \int P_{\omega,s,a} b_s^y(d\omega)$$

and conditional losses

$$\ell_y^x(s, a) := \int \ell_\omega(s, a) b_s^y(d\omega)$$

are determined by s .

The quotient of histories by the objects in Definition 2.3 is the minimal exact representation. Approximate representations are handled by adding approximation errors to the Bellman inequalities. A DMSO model class (Foster et al., 2021) is one possible instantiation of the kernels and losses, but it is not required for, nor intrinsic to, tight upper and lower bounds in indexed sequential experiments. The true requirement is Bellman sufficiency: the state must know enough to predict, update, evaluate loss, and compute information. This is aligned with the representational philosophy behind low Bellman rank for contextual decision processes (Jiang et al., 2017) and admissible relaxations in online learning (Rakhlin et al., 2012).

3 Indexed information and exact AIR/MAIR identities

3.1 Posterior-reference histories and index information

For a prior μ and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}_\mu^{\text{Alg}} := \int \mathbb{P}_\omega^{\text{Alg}} \mu(d\omega).$$

A reference history is denoted

$$H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}, \quad S'_t = \phi_t(H'_{t-1}, b'_t), \quad b'_t = \mathcal{L}_\mu(\Omega \mid H'_{t-1}),$$

and the algorithm's reference decision kernel is

$$p'_t = p_t(\cdot \mid H'_{t-1}).$$

Definition 3.1 (Indexed information gain). *At state s , let q_s be the posterior law of $Y = \chi(\Omega)$ and let $P_{s,a}^y$ and $P_{s,a}$ be the conditional and marginal predictive laws from Definition 2.3. For $p \in \Delta(\mathcal{A}_t)$, define*

$$\mathcal{I}_\chi(s, p) := \mathbb{E}_{a \sim p} \int D_{\text{KL}}(P_{s,a}^y \parallel P_{s,a}) q_s(dy). \quad (4)$$

When $Y = \Omega$, this is model-index information. When $Y = \pi^(\Omega)$, this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.*

Define the cumulative and average indexed information of an algorithm by

$$C_{\chi}^{\text{Alg}}(\mu) := \mathbb{E}_{H'_T \sim \mathbb{P}_{\mu}^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_{\chi}(S'_t, p'_t), \quad \bar{C}_{\chi}^{\text{Alg}}(\mu) := \frac{1}{T} C_{\chi}^{\text{Alg}}(\mu). \quad (5)$$

Proposition 3.2 (Reference-history indexed chain rule). *For every prior μ , index map χ , and algorithm Alg ,*

$$I_{\mu}(Y; H_T) = \mathbb{E}_{H'_T \sim \mathbb{P}_{\mu}^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_{\chi}(S'_t, p'_t) = T \bar{C}_{\chi}^{\text{Alg}}(\mu). \quad (6)$$

Proof. The mutual-information chain rule gives

$$I_{\mu}(Y; H_T) = \sum_{t=1}^T I_{\mu}(Y; A_t, O_t \mid H_{t-1}).$$

Given H_{t-1} , the action A_t is sampled by the algorithm and is conditionally independent of Y . Therefore

$$I_{\mu}(Y; A_t, O_t \mid H_{t-1}) = I_{\mu}(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on $A_t = a$, the law of O_t given $Y = y$ is $P_{S_t, a}^y$, while the posterior predictive law is $P_{S_t, a}$. Hence

$$I_{\mu}(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\text{KL}}(P_{S_t, a}^y \parallel P_{S_t, a}) q_{S_t}(dy).$$

Averaging over $a \sim p_t(\cdot \mid H_{t-1})$ and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression. $\square$

3.2 Fixed-truth log gain and deterministic specialization

The information gain in (4) is posterior-averaged. The regret identity used in frequentist upper bounds is slightly finer: it fixes the true environment and tracks the posterior log ratio of its index.

Fix a state s , an environment ω , and let $y = \chi(\omega)$. Assume domination and $q_s(y) > 0$. Define

$$\Lambda_{\chi}(s, a, o; y) := \log \frac{dP_{s, a}^y}{dP_{s, a}}(o).$$

For $p \in \Delta(\mathcal{A})$, define the fixed-truth index log gain

$$J_{\chi}(s, p; \omega) := \mathbb{E}_{a \sim p, O \sim P_{\omega, s, a}} \Lambda_{\chi}(s, a, O; \chi(\omega)). \quad (7)$$

Equivalently,

$$J_{\chi}(s, p; \omega) = \mathbb{E}_{a \sim p} \left[D_{\text{KL}}(P_{\omega, s, a} \parallel P_{s, a}) - D_{\text{KL}}(P_{\omega, s, a} \parallel P_{s, a}^{\chi(\omega)}) \right]. \quad (8)$$

Thus the fixed-truth log gain equals the indexed information only after posterior averaging over ω conditional on the index, or in special cases in which the true conditional law already equals the conditional predictive component. In general, the residual

$$D_{\text{KL}}(P_{\omega, s, a} \parallel P_{s, a}^{\chi(\omega)})$$

is a calibration term. This is why UCB proofs need both self-normalized calibration and optimism: optimism turns regret into a width, and calibration turns the fixed truth into the posterior-reference information geometry.

For $\eta > 0$, define the fixed-truth indexed bracket

$$B_{\chi,\eta}(s, p; \omega) := \ell_\omega(s, p) - \frac{1}{\eta} J_\chi(s, p; \omega), \quad \ell_\omega(s, p) := \mathbb{E}_{a \sim p} \ell_\omega(s, a). \quad (9)$$

Corollary 3.3 (Deterministic-action indexed regret identity). *Suppose the algorithm is deterministic after conditioning on H_{t-1} , so $p_t = \delta_{a_t}$. Fix ω^* and $y^* = \chi(\omega^*)$. If the reference index posterior is updated by Bayes' rule, then conditionally on H_{t-1} ,*

$$\ell_{\omega^*}(S_t, a_t) = B_{\chi,\eta}(S_t, \delta_{a_t}; \omega^*) + \frac{1}{\eta} \mathbb{E}_{O_t \sim P_{\omega^*, S_t, a_t}} \left[\log \frac{q_{t+1}(y^*)}{q_t(y^*)} \middle| H_{t-1} \right]. \quad (10)$$

Consequently,

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) = \frac{1}{\eta} \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{q_{T+1}(y^*)}{q_1(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T B_{\chi,\eta}(S_t, \delta_{a_t}; \omega^*). \quad (11)$$

The outer expectation is only over the observation process and any exogenous randomness. In a deterministic noiseless experiment the same identity is pathwise.

Proof. Bayes' rule gives

$$\log \frac{q_{t+1}(y^*)}{q_t(y^*)} = \log \frac{dP_{S_t, a_t}^{y^*}}{dP_{S_t, a_t}}(O_t).$$

Taking conditional expectation under the fixed truth ω^* gives $J_\chi(S_t, \delta_{a_t}; \omega^*)$. Rearranging the definition of $B_{\chi,\eta}$ proves the one-step identity. Summing over t telescopes the posterior log ratio. $\square$

This corollary is the deterministic specialization requested in UCB analyses. It drops the outer expectation over randomized decision laws; what remains is the posterior-ratio telescope and the fixed-truth gradient bracket.

3.3 Exact AIR and MAIR regret identities

We now record the finite form used in the AIR/MAIR literature (Xu and Zeevi, 2025). Although we use the DMSO language to state the AIR/MAIR results and facilitate comparison with earlier work, the results in this subsection require only the Bellman-sufficient representation of Definition 2.3. There, the index may be either a well-specified model or an optimal decision in the misspecified/adversarial setting, while the state is simply the realized history.

Let $\mathcal{M}$ be the model class and $\mathcal{A}$ be the decision space. Model M has reward mean $r_M(a)$ and observation law $P_{M,a}$. For a candidate optimal action a^* , write

$$\Delta_{M,a^*}(p) := r_M(a^*) - \mathbb{E}_{a \sim p} r_M(a).$$

AIR. An AIR belief is a pair belief $\nu \in \Delta(\mathcal{M} \times \mathcal{A})$ over (M, A^*) , while the reference is an action marginal $q \in \text{int } \Delta(\mathcal{A})$. Define

$$\text{AIR}_{q,\eta}(p, \nu) := \mathbb{E}_{(M, A^*) \sim \nu} \Delta_{M,A^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} I_\nu(A^*; O \mid a) - \frac{1}{\eta} D_{\text{KL}}(q_\nu \| q), \quad (12)$$

where q_ν is the A^* -marginal of ν . Equivalently,

$$\text{AIR}_{q,\eta}(p, \nu) = \mathbb{E}_{(M, A^*), a, O} \left[\Delta_{M,A^*}(\delta_a) - \frac{1}{\eta} \log \frac{\nu(A^* \mid a, O)}{q(A^*)} \right].$$

Lemma 3.4 (AIR gradient bracket). *Assume finite spaces, or dominated spaces where the displayed Gateaux derivatives exist. Up to an additive constant on the simplex,*

$$\frac{\partial}{\partial \nu(M, a^*)} \text{AIR}_{q,\eta}(p, \nu) = \Delta_{M, a^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p, O \sim P_{M,a}} \log \frac{\nu(A^* = a^* \mid a, O)}{q(a^*)}. \quad (13)$$

Consequently, for every fixed pair (M, a^*) ,

$$B_{q,\eta}^{\text{AIR}}(p, \nu; M, a^*) := \text{AIR}_{q,\eta}(p, \nu) + \langle \nabla_\nu \text{AIR}_{q,\eta}(p, \nu), \delta_{(M, a^*)} - \nu \rangle \quad (14)$$

$$= \Delta_{M, a^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p, O \sim P_{M,a}} \log \frac{\nu(A^* = a^* \mid a, O)}{q(a^*)}. \quad (15)$$

Proof. For fixed a , set

$$w_{a^*, o} = \sum_M \nu(M, a^*) P_{M,a}(o), \quad w_o = \sum_{a^*} w_{a^*, o}.$$

Then

$$I_\nu(A^*; O \mid a) + D_{\text{KL}}(q_\nu \| q) = \sum_{a^*, o} w_{a^*, o} \log \frac{w_{a^*, o}}{w_o q(a^*)}.$$

Differentiating with respect to $\nu(M, a^*)$ gives

$$\mathbb{E}_{O \sim P_{M,a}} \log \frac{\nu(A^* = a^* \mid a, O)}{q(a^*)},$$

because the $+1$ terms from differentiating $w_{a^*, o} \log w_{a^*, o}$ and $w_o \log w_o$ cancel. Adding the derivative of the linear regret term proves (13). Pairing the gradient with $\delta_{(M^*, a^*)} - \nu$ cancels the posterior-averaged terms and gives (14). $\square$

Theorem 3.5 (Exact AIR regret identity). *Let an algorithm generate decision laws p_t , pair beliefs ν_t , and reference action beliefs $q_t \in \text{int } \Delta(\mathcal{A})$. After action a_t and observation O_t , set*

$$q_{t+1}(\cdot) = \nu_t(A^* \in \cdot \mid a_t, O_t).$$

For an adversarial model sequence $\mathbf{M} = (M_1, \dots, M_T) \in \mathcal{M}^T$ with best-in-hindsight action a^ ; we omit this case only to keep the notation simple. Then, for every $\eta > 0$,*

$$\text{Reg}_T(\text{Alg}, \mathbf{M}) = \frac{1}{\eta} \mathbb{E} \log \frac{q_{T+1}(a^*)}{q_1(a^*)} + \mathbb{E} \sum_{t=1}^T B_{q_t, \eta}^{\text{AIR}}(p_t, \nu_t; M^*, a^*). \quad (16)$$

Proof. Condition on H_{t-1} . By the definition of the action-marginal posterior,

$$\mathbb{E} \left[\log \frac{q_{t+1}(a^*)}{q_t(a^*)} \mid H_{t-1} \right] = \mathbb{E}_{a \sim p_t, O \sim P_{M^*, a}} \log \frac{\nu_t(A^* = a^* \mid a, O)}{q_t(a^*)}.$$

Combining this equality with (14) gives a one-step equality between regret, bracket, and posterior-ratio increment. Taking expectations and summing over time telescopes the logarithm. $\square$

MAIR. For MAIR the belief is $\mu \in \Delta(\mathcal{M})$ and the reference is $\rho \in \text{int } \Delta(\mathcal{M})$. Write $\mu_{O|a} = \int P_{M,a} \mu(dM)$ and

$$\Delta_M(p) = r_M(a_M^*) - \mathbb{E}_{a \sim p} r_M(a), \quad a_M^* \in \arg \max_a r_M(a).$$

Define

$$\text{MAIR}_{\rho,\eta}(p, \mu) := \mathbb{E}_{M \sim \mu} \Delta_M(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} I_\mu(M; O | a) - \frac{1}{\eta} D_{\text{KL}}(\mu \| \rho). \quad (17)$$

The same differentiation gives

$$B_{\rho,\eta}^{\text{MAIR}}(p, \mu; M^*) := \text{MAIR}_{\rho,\eta}(p, \mu) + \langle \nabla_\mu \text{MAIR}_{\rho,\eta}(p, \mu), \delta_{M^*} - \mu \rangle \quad (18)$$

$$= \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{M^*,a} \| \mu_{O|a}) - \frac{1}{\eta} \log \frac{\mu(M^*)}{\rho(M^*)}. \quad (19)$$

If $\rho_{t+1}(\cdot) = \mu_t(\cdot | a_t, O_t)$, then

$$\text{Reg}_T(\text{Alg}, M^*) = \frac{1}{\eta} \mathbb{E} \log \frac{\rho_{T+1}(M^*)}{\rho_1(M^*)} + \mathbb{E} \sum_{t=1}^T B_{\rho_t,\eta}^{\text{MAIR}}(p_t, \mu_t; M^*). \quad (20)$$

The proof is the same posterior-ratio telescope as for AIR. Thus AIR and MAIR are one indexed identity with different choices of Y .

Remark 3.6 (Danskin, Nash equilibrium, and why the bracket is not accidental). *The direct proofs above are intentionally short. The original AIR proof of Xu and Zeevi (2025, Section 5) gives the variational reason behind the same equations. Introduce an auxiliary prediction rule $Q : (a, o) \mapsto \Delta(\mathcal{A})$ and the posterior-scoring objective*

$$\mathbb{E}_{(M,A^*),a,O} \left[\Delta_{M,A^*}(\delta_a) - \frac{1}{\eta} \log \frac{Q[a,O](A^*)}{q(A^*)} \right].$$

For fixed ν , the logarithmic score is strongly convex in the predictive distribution and is uniquely minimized by the Bayesian posterior $Q_\nu[a, o] = \nu(A^ \in \cdot | a, o)$. The optimized value is the AIR objective; the unoptimized objective is linear, hence concave, in ν . Danskin's theorem identifies the AIR gradient with the derivative at the optimizer Q_ν . This is the conceptual content of Lemma 5.3 in Xu and Zeevi (2025). The same concavity plus strong convexity also gives the Nash equilibrium statement of Lemma 5.2 by first-order/Danskin reasoning, without invoking a separate fixed-point theorem. This does not necessarily simplify the full proof, because compactness, differentiability, and boundary conditions still have to be checked. It does explain why the gradient bracket is not an accidental cancellation: posterior scoring produces the minimizer, Danskin produces the gradient, and the regret identity telescopes the posterior reference.*

The perspective developed in Xu and Zeevi (2025) is that frequentist sequential learning can be organized around algorithmic beliefs μ , interpreted as predictive laws, and decision rules induced by reference or posterior beliefs ρ , either through posterior sampling or through robust minimax optimization. We use MAIR to illustrate how a single functional captures both the confidence multiplier in UCB-type algorithms and the decision-estimation coefficient in one-round minimax rules.

Lemma 3.7 (MAIR gradient bracket and three canonical specializations). *Assume the observation laws are dominated and the displayed derivatives exist. For fixed p, ρ, μ , up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \mu(M)} \text{MAIR}_{\rho, \eta}(p, \mu) = \Delta_M(p) - \frac{1}{\eta} \log \frac{\mu(M)}{\rho(M)} - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M, \pi} \| \mu_{o|\pi}).$$

Consequently, for every M^* ,

$$\begin{aligned} & \text{MAIR}_{\rho, \eta}(p, \mu) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \mu), \delta_{M^*} - \mu \rangle \\ &= \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M^*, \pi} \| \mu_{o|\pi}) - \frac{1}{\eta} \log \frac{\mu(M^*)}{\rho(M^*)}. \end{aligned} \quad (21)$$

This bracket has three important specializations.

1. If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset

$$\Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M^*, \pi} \| \mu_{o|\pi}). \quad (22)$$

In Section 3.4 we will show that for GP-UCB,

$$\text{self-normalized calibration} + \text{UCB optimism} \Rightarrow \text{MAIR gradient-error condition}.$$

2. If $\bar{\mu} \in \arg \max_{\mu \in \Delta(\mathcal{M})} \text{MAIR}_{\rho, \eta}(p, \mu)$ is an interior maximizer, then

$$\text{MAIR}_{\rho, \eta}(p, \bar{\mu}) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \bar{\mu}), \delta_{M^*} - \bar{\mu} \rangle \leq \text{MAIR}_{\rho, \eta}(p, \bar{\mu}). \quad (23)$$

This is the MAMS/MEBO bracket control.

3. Let $p_{M, \pi} = dP_{M, \pi} / d\nu_{\pi}$ and define, for an auxiliary pair $(\bar{\pi}, \bar{o})$,

$$\frac{d\mu_{\bar{\pi}, \bar{o}}^{1/2}}{d\rho}(M) := \frac{\sqrt{p_{M, \bar{\pi}}(\bar{o})}}{\int \sqrt{p_{N, \bar{\pi}}(\bar{o})} \rho(dN)}. \quad (24)$$

Let $h^2(P, Q) := 1 - \int \sqrt{dP dQ}$. If $\bar{\pi} \sim p$ and $\bar{o} \sim P_{M^*, \bar{\pi}}$, then

$$\begin{aligned} & \mathbb{E}_{\bar{\pi}, \bar{o}} \left[\text{MAIR}_{\rho, \eta}(p, \mu_{\bar{\pi}, \bar{o}}^{1/2}) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \mu_{\bar{\pi}, \bar{o}}^{1/2}), \delta_{M^*} - \mu_{\bar{\pi}, \bar{o}}^{1/2} \rangle \right] \\ & \leq \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} \mathbb{E}_{M \sim \rho} h^2(P_{M^*, \pi}, P_{M, \pi}). \end{aligned} \quad (25)$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger E2D/DEC offset.

Proof. The derivative calculation is direct. The derivative of $\mathbb{E}_{\mu} \Delta_M(p)$ is $\Delta_M(p)$; the derivative of $D_{\text{KL}}(\mu \| \rho)$ is $\log(\mu(M)/\rho(M)) + 1$; and differentiating the predictive mixture in the mutual-information term gives $D_{\text{KL}}(P_{M, \pi} \| \mu_{o|\pi})$, up to the same simplex constant. Pairing the derivative with $\delta_{M^*} - \mu$ cancels the averaged terms in (17) and proves (21).

The case $\mu = \rho$ is immediate from the logarithmic cancellation.

The case $\bar{\mu} \in \arg \max \text{MAIR}$ is from the first-order optimality.

It remains to justify the square-root claim; this is a standard fact for the squared Hellinger distance. For fixed $\bar{\pi}$, let

$$Z_{\bar{\pi}}(\bar{o}) := \int \sqrt{p_{N, \bar{\pi}}(\bar{o})} \rho(dN).$$

Then

$$\log \frac{d\mu_{\bar{\pi}, \bar{o}}^{1/2}}{d\rho}(M^\star) = \frac{1}{2} \log p_{M^\star, \bar{\pi}}(\bar{o}) - \log Z_{\bar{\pi}}(\bar{o}).$$

Under $\bar{o} \sim P_{M^\star, \bar{\pi}}$, Jensen's inequality gives

$$\begin{aligned} \mathbb{E} \log \frac{d\mu_{\bar{\pi}, \bar{o}}^{1/2}}{d\rho}(M^\star) &= -\mathbb{E} \log \frac{Z_{\bar{\pi}}(\bar{o})}{\sqrt{p_{M^\star, \bar{\pi}}(\bar{o})}} \\ &\geq -\log \int \sqrt{p_{M^\star, \bar{\pi}}(o)} Z_{\bar{\pi}}(o) d\nu_{\bar{\pi}}(o) \\ &= -\log (1 - \mathbb{E}_{M \sim \rho} h^2(P_{M^\star, \bar{\pi}}, P_{M, \bar{\pi}})) \\ &\geq \mathbb{E}_{M \sim \rho} h^2(P_{M^\star, \bar{\pi}}, P_{M, \bar{\pi}}). \end{aligned}$$

Averaging over $\bar{\pi} \sim p$, substituting into (21), and dropping the nonnegative KL term proves (25). $\square$

3.4 Upper bounds: one identity and three algorithm families

We firstly give a generic indexed-information supersolution.

Let Φ_t be any predictable potential on states. For every algorithm, the algebraic identity

$$\mathbb{E} L_\Omega(H_T) = \mathbb{E} \sum_{t=1}^T [\ell(S_t, p_t) + \mathbb{E}[\Phi_{t+1}(S_{t+1}) \mid S_t, p_t] - \Phi_t(S_t)] \quad (26)$$

$$+ \Phi_1(s_1) - \mathbb{E} \Phi_{T+1}(S_{T+1}) \quad (27)$$

holds by telescoping.

Theorem 3.8 (Generic indexed AIR/MAIR upper certificate). *Suppose an algorithm chooses p_t such that for all reached states,*

$$\ell(S_t, p_t) + \mathbb{E}[\Phi_{t+1}(S_{t+1}) \mid S_t, p_t] - \Phi_t(S_t) \leq \gamma_t \mathcal{I}_\chi(S_t, p_t) + \varepsilon_t. \quad (28)$$

Then

$$\mathbb{E} L_\Omega(H_T) \leq \Phi_1(s_1) - \mathbb{E} \Phi_{T+1}(S_{T+1}) + \mathbb{E} \sum_{t=1}^T \gamma_t \mathcal{I}_\chi(S_t, p_t) + \sum_{t=1}^T \varepsilon_t. \quad (29)$$

If $\gamma_t \leq \gamma$ and $\Phi_{T+1} \geq 0$, then

$$\mathbb{E} L_\Omega(H_T) \leq \Phi_1(s_1) + \gamma I_\mu(Y; H_T) + \sum_t \varepsilon_t.$$

Proof. Substitute (28) into (26). The last display uses Proposition 3.2. $\square$

The bracket in (28) is the common object. UCB, E2D/DEC, and AMS/EBO differ in how they control it. In the model-index specialization, write the posterior/reference belief as μ_t and define the fixed-truth MAIR information increment

$$J_t^{\text{MAIR}}(p; \omega^\star) := \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^\star, t}(\cdot \mid S_t, a) \parallel P_{\mu_t, t}(\cdot \mid S_t, a)). \quad (30)$$

The MAIR gradient bracket is the quantity obtained from the left-hand side of (28) after subtracting $\eta^{-1} J_t^{\text{MAIR}}(p; \omega^\star)$, together with the posterior-ratio term in (20). The action-index AIR version is identical with the index $Y = A^\star$ and the conditional predictive laws $P_{t, y, a}$ in place of $P_{\omega, t, a}$. Thus the same theorem may be read either as an AIR statement about learning the optimal decision or as a MAIR statement about estimating the model/environment.

1. UCB families: calibration plus optimism. An upper-confidence-bound (UCB) algorithm maintains confidence sets or posterior predictive bands and chooses an action that is optimistic within that representation (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). Calibration says the fixed truth lies inside the represented uncertainty set. Optimism converts regret into a posterior width. A self-normalized martingale inequality or elliptical-potential/log-determinant lemma then sums the squared widths into a realized information telescope. In this language, UCB does not solve the AIR/MAIR bracket optimization; it certifies the bracket after the action is chosen.

Lemma 3.9 (Self-normalized optimism implies bracket control). *Fix a truth ω^* and suppose that on an event $\mathcal{E}$, for every reached round t there exist a predictable width $w_t(a) \geq 0$, a tractable information increment $g_t \geq 0$, and constants $c_1, c_2 > 0$ such that*

$$\ell_{\omega^*}(S_t, p_t) \leq c_1 \beta_t \left(\mathbb{E}_{a \sim p_t} w_t(a)^2 \right)^{1/2}, \quad \mathbb{E}_{a \sim p_t} w_t(a)^2 \leq c_2 g_t. \quad (31)$$

Then for every $\eta > 0$,

$$\ell_{\omega^*}(S_t, p_t) - \frac{1}{\eta} g_t \leq \frac{\eta c_1^2 c_2 \beta_t^2}{4}. \quad (32)$$

If the exact fixed-truth indexed bracket uses $J_{\chi, t} := J_{\chi}(S_t, p_t; \omega^*)$, then

$$B_{\chi, \eta}(S_t, p_t; \omega^*) \leq \frac{\eta c_1^2 c_2 \beta_t^2}{4} + \frac{1}{\eta} (g_t - J_{\chi, t}). \quad (33)$$

Proof. Deterministic-action indexed regret identity follows the Corollary 3.3. By (31),

$$\ell_{\omega^*}(S_t, p_t) \leq c_1 \beta_t \sqrt{c_2 g_t}.$$

Young’s inequality $ab \leq \eta a^2/4 + b^2/\eta$ gives (32). Subtracting $\eta^{-1} J_{\chi, t}$ instead of $\eta^{-1} g_t$ gives (33). $\square$

The correction $g_t - J_{\chi, t}$ is not a second estimation cost. It is the algebraic difference between the fixed-truth posterior log gain in the gradient bracket and the tractable reference information increment. In linear-UCB and GP-UCB proofs this difference either cancels against the posterior-ratio telescope or is bounded by the same determinant/elliptical-potential term. Therefore

$$\text{self-normalized calibration} + \text{optimism} \implies \text{AIR/MAIR gradient-bracket control}.$$

A central organizing perspective of this paper is that the UCB proof can be read as a MAIR decomposition. The round-heterogeneous summation of posterior widths, usually controlled by the elliptical-potential or log-determinant argument, plays the role of a telescoping estimation-complexity term: it is the accumulated algorithmic information or fixed-truth estimation loss. The confidence multiplier, by contrast, enters as a per-round upper bound on the MAIR/DEC bracket, i.e., on the coefficient that converts posterior information into regret. This interpretation may be a bit counterintuitive at first sight from the finite-action UCB proof viewpoint: what is often called the “estimation” difficulty in self-normalized martingale concentration becomes, in MAIR language, part of the regret-to-information coefficient; while the elliptical-potential summation becomes the estimation-complexity telescope. This reversal is precisely the model-based view: posterior widths quantify the “estimation complexity” of the underlying parameter, as explained by Lattimore (2023). The resulting unification is therefore not merely an algebraic rearrangement of known facts. It reveals a common round-heterogeneous posterior-telescoping structure while separating the distinct estimation and coefficient mechanisms behind UCB, E2D, and AMS/EBO.

2. E2D/DEC: greedy MAIR offset optimization. The decision-estimation coefficient (DEC) (Foster et al., 2021) and estimation-to-decision (E2D) algorithms choose p by optimizing a one-step Lagrangian relaxation. In model-index notation this takes the schematic form

$$\inf_{p \in \Delta(\mathcal{A})} \sup_{\omega \in \mathcal{N}(s)} \{ \ell_\omega(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega, s, a} \| P_{\rho, s, a}) \}, \quad (34)$$

where ρ is a reference belief or reference model and $\mathcal{N}(s)$ is a localized comparison set. The same one-round minimax optimization principle also applies to the original Hellinger form and constrained variants (Foster et al., 2023). Equation (34) is exactly the MAIR gradient bracket without the Bellman continuation term

$$\mathbb{E} \Phi_{t+1}(\tau(s, A, O)) - \Phi_t(s).$$

Thus E2D/DEC is a greedy relaxation of the indexed Bellman program. It can be sharp when the posterior geometry is simple and the one-step witnesses paste together dynamically. Its role with realized information gain is subtler: an unlocalized DEC may be infinite or vacuous, while a localized DEC must be compatible with the same posterior-reference telescope used by the upper or lower proof. Otherwise one risks combining a coefficient that already contains estimation difficulty with a separate log-determinant telescope.

3. AMS/EBO: robust saddle control over the indexed bracket. Adaptive model selection or exploration-by-optimization methods, including AMS/EBO and mirror-descent information-ratio algorithms (Lattimore and György, 2021; Xu and Zeevi, 2025), control a robust saddle version of the same bracket. A schematic form is

$$\inf_p \sup_{\nu \in \mathcal{U}(s)} \{ \ell_\nu(s, p) + \mathbb{E} V_{t+1}(S') - V_t(s) - \gamma \mathcal{I}_{\chi, \nu}(s, p) \}.$$

In the AIR specialization $\chi(\Omega) = A^*(\Omega)$, the sequential information telescope is at most $H(A^*)$. For a finite active action set this is at most $\log |\mathcal{A}|$, and for a low-dimensional action index it is the corresponding index entropy. Consequently AMS/EBO has a natural route to finite-action or low-dimensional action-set rates: it can scale with the decision-index complexity and the sequential AIR bracket, rather than with the ambient model dimension. Converting this route into a fixed-truth frequentist theorem still requires calibration of the algorithmic belief, but the information accounting is representation-level and does not require a global one-round DEC.

Remark 3.10 (The one identity). *The identity is the potential telescope (26), refined by the AIR/MAIR posterior-ratio identities. UCB certifies the bracket through confidence and optimism. E2D/DEC greedily optimizes the MAIR/DEC offset. AMS/EBO controls a robust saddle over indexed beliefs. The algorithms differ, but they act on the same algebraic decomposition of regret into a potential telescope plus an indexed-information bracket.*

4 Lower bounds: reference histories and quantile indices

4.1 Ghost probability and true good probability

Fix an average-regret threshold $r \in [0, 1]$. The posterior-reference ghost-good probability is

$$p_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y \sim \mu_\chi, H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} (\bar{L}_\chi(Y, H'_T) \leq r), \quad (35)$$

where Y and H'_T are independent under the ghost law and μ_χ is the marginal law of Y . The true good probability is

$$q_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y, H_T} (\bar{L}_\chi(Y, H_T) \leq r), \quad (36)$$

where (Y, H_T) are generated by $\Omega \sim \mu$ and $H_T \sim \mathbb{P}_\Omega^{\text{Alg}}$.

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

4.2 Reference-history quantile theorem

Let

$$\text{kl}(q, p) = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}$$

be binary KL divergence. For $p \in [0, 1]$ and $c \geq 0$, define

$$\psi(p, c) := \sup\{q \in [0, 1] : \text{kl}(q, p) \leq c\}.$$

Theorem 4.1 (Reference-history quantile indexed-information lower bound). *For every prior μ , index map χ , threshold $r \in [0, 1]$, and algorithm Alg ,*

$$\text{kl}(q_r^{\text{Alg}}(\mu, \chi), p_r^{\text{Alg}}(\mu, \chi)) \leq C_\chi^{\text{Alg}}(\mu) = T\bar{C}_\chi^{\text{Alg}}(\mu). \quad (37)$$

Consequently,

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[1 - \psi(p_r^{\text{Alg}}(\mu, \chi), T\bar{C}_\chi^{\text{Alg}}(\mu)) \right]. \quad (38)$$

In particular, if

$$T\bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad (39)$$

then

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} \bar{L}_\Omega(H_T) \geq \frac{r}{2}, \quad \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (40)$$

A convenient sufficient condition is

$$T\bar{C}_\chi^{\text{Alg}}(\mu) + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r^{\text{Alg}}(\mu, \chi)}. \quad (41)$$

Proof. Consider the true joint law of (Y, H_T) and the ghost law of (Y, H'_T) . Both have the same marginal law of Y , while Y and H'_T are independent under the ghost law. Hence

$$D_{\text{KL}}(\mathcal{L}(Y, H_T) \| \mathcal{L}(Y, H'_T)) = I_\mu(Y; H_T).$$

Data processing applied to the indicator of the event $\{\bar{L}_\chi(Y, \cdot) \leq r\}$ gives

$$\text{kl}(q_r^{\text{Alg}}, p_r^{\text{Alg}}) \leq I_\mu(Y; H_T).$$

Proposition 3.2 identifies the right-hand side with $C_\chi^{\text{Alg}}(\mu) = T\bar{C}_\chi^{\text{Alg}}(\mu)$, proving (37). The definition of ψ gives $q_r^{\text{Alg}} \leq \psi(p_r^{\text{Alg}}, T\bar{C}_\chi^{\text{Alg}})$. By nonnegativity of loss and by (3),

$$\begin{aligned} \mathbb{E}_{\Omega, H_T} L_\Omega(H_T) &= \mathbb{E}_{Y, H_T} L_\chi(Y, H_T) \\ &\geq Tr \mathbb{P}\{\bar{L}_\chi(Y, H_T) > r\} \\ &= Tr(1 - q_r^{\text{Alg}}), \end{aligned}$$

which proves (38). If (39) holds, then $q_r^{\text{Alg}} \leq 1/2$, giving (40). Finally,

$$\text{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2} \log \frac{1}{4p(1-p)} \geq \frac{1}{2} \log \frac{1}{p} - \log 2,$$

so (41) implies (39). $\square$

Relation to interactive Fano. Theorem 4.1 should be viewed as an indexed, posterior-reference specialization of the interactive Fano method of Chen et al. (2024). The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index $Y = \chi(\Omega)$. With this choice, the trajectory information admits the exact stepwise identity

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(b'_t, p'_t).$$

Thus the same reference history H'_T generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular, $\chi(\Omega) = \Omega$ gives the model-index MAIR form, while $\chi(\Omega) = \pi^*(\Omega)$ gives the action-index AIR form.

The novelty is not a new Fano inequality, but a new representation of interactive Fano in posterior-reference indexed-information coordinates. This avoids prematurely collapsing the adaptive experiment to a one-round decision distribution, the step that creates the localized change-of-measure difficulty in one-round DEC reductions. It also makes the lower-bound critical radius directly comparable to AIR/MAIR supersolution upper bounds, since both sides are expressed as sums of the same indexed information along posterior-reference histories.

Remark 4.2 (Why average notation is useful). *The theorem could be written entirely with cumulative quantities. Writing $\bar{C}_\chi = T^{-1}\mathbb{E} \sum_t \mathcal{I}_\chi$ and $\bar{L} = T^{-1}L$ makes the scaling visible:*

$$\text{information side} = T\bar{C}_\chi, \quad \text{regret side} = Tr.$$

Thus T appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.

4.3 Bellman-Fano lower certificates

Theorem 4.1 is algorithm-specific. We now give algorithm-uniform certificates. The first bounds the information of any algorithm; the second bounds the ghost probability of success of any algorithm.

Information supersolution. For a function F on states, define

$$(\mathbf{C}_t F)(s) := \sup_{p \in \Delta(\mathcal{A}_t)} \left\{ \mathcal{I}_\chi(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s,a}} F(\tau(s, a, o)) \right\}.$$

A sequence $\bar{C}_t$ is an information supersolution if

$$\bar{C}_{T+1} \geq 0, \quad \bar{C}_t(s) \geq (\mathbf{C}_t \bar{C}_{t+1})(s) \quad \forall t, s.$$

Then, for every algorithm starting from s_1 ,

$$C_\chi^{\text{Alg}}(\mu) \leq \bar{C}_1(s_1). \tag{42}$$

The proof is the same backward induction as Proposition 2.2.

Ghost-good-mass supersolution. Let $m \in \Delta(\mathcal{Y})$ be the target-index distribution used to evaluate success, let s be the posterior-reference state, and let $g : \mathcal{Y} \rightarrow [0, 1]$ be an accumulated average-loss profile. For threshold r , define the exact recursion

$$\Gamma_{T+1}^r(m, s, g) = m\{y : g(y) \leq r\},$$

$$\Gamma_t^r(m, s, g) = \sup_{p \in \Delta(\mathcal{A}_t)} \mathbb{E}_{a \sim p, o \sim P_{s,a}} \Gamma_{t+1}^r \left(m, \tau(s, a, o), g + \frac{1}{T} \ell^X(s, a) \right),$$

where $\ell^X(s, a)$ is the function $y \mapsto \ell_y^X(s, a)$. If $\bar{\Gamma}_t^r$ is a supersolution of this recursion, then for every algorithm,

$$p_r^{\text{Alg}}(\mu, \chi) \leq \bar{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (43)$$

Define the ghost entropy lower certificate

$$\underline{\mathcal{E}}_1^r(\mu, \chi) := -\log \bar{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (44)$$

Theorem 4.3 (Bellman quantile-index lower certificate). *Suppose $\bar{C}_t$ is an information supersolution and $\bar{\Gamma}_t^r$ is a ghost-good-mass supersolution. Then*

$$\inf_{\text{Alg}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[1 - \psi(e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}, \bar{C}_1(s_1)) \right]. \quad (45)$$

In particular, if

$$\bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi), \quad (46)$$

then

$$\inf_{\text{Alg}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (47)$$

Consequently,

$$\mathfrak{R}_T^* \geq \sup_{\mu, r} \left\{ \frac{Tr}{2} : \bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}.$$

Proof. For every algorithm, (42) gives $T\bar{C}_\chi^{\text{Alg}}(\mu) = C_\chi^{\text{Alg}}(\mu) \leq \bar{C}_1(s_1)$, and (43) gives $p_r^{\text{Alg}}(\mu, \chi) \leq e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}$. Substitute these two inequalities into Theorem 4.1. The minimax statement follows from Yao's principle. $\square$

The critical average-regret radius is

$$r_*(\mu, \chi) := \sup \left\{ r : \bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}. \quad (48)$$

The lower bound is order Tr_* . A two-point Le Cam proof corresponds to the special case in which $\underline{\mathcal{E}} = O(1)$. A Fano or local-prior-mass lower bound has $\underline{\mathcal{E}}$ of order dimension or effective dimension.

4.4 DEC as one-step greedy relaxation

At a state s , let ν be a local prior over environments compatible with the state. The local indexed information is

$$\mathcal{I}_\chi(s, p; \nu) = \mathbb{E}_{a \sim p} I_\nu(\chi(\Omega); O \mid s, a),$$

and the local average loss is

$$\ell_\nu(s, p) = \mathbb{E}_{\omega \sim \nu, a \sim p} \ell_\omega(s, a).$$

A one-step offset coefficient has the form

$$\mathfrak{d}_\chi(s, \gamma) := \inf_{p \in \Delta(\mathcal{A})} \sup_{\nu \in \mathcal{N}(s)} \{\ell_\nu(s, p) - \gamma \mathcal{I}_\chi(s, p; \nu)\}.$$

A quantile version replaces $\ell_\nu(s, p)$ by an upper quantile of $\ell_\Omega(s, A)$ under $\Omega \sim \nu$ and $A \sim p$. These coefficients resemble the upper bracket and are useful for algorithm design. The lower-bound caveat is dynamic consistency: a free per-state $\sup_\nu$ may choose witnesses that cannot be pasted into one posterior process. A valid stochastic minimax lower bound needs either an actual prior and posterior-reference history, as in Theorem 4.1, or a Bellman certificate, as in Theorem 4.3.

Thus DEC/E2D should be read as a one-step, greedy relaxation of indexed Bellman information complexity, rather than as a universally tight conversion coefficient from information to regret. When the posterior geometry is simple and essentially one-round, this relaxation can be sharp. In more structured problems, however, a one-round coefficient can obscure the dynamic object in two distinct ways: it may charge an estimation or entropy cost that the posterior-reference lower bound has already paid, or it may discard the geometry of the reference trajectory that makes localization possible. The examples in Section 5 give tight indexed-Bellman certificates; the following informal diagnostics explain why the corresponding one-round DEC accounting must be interpreted with care.

- *Vacuity for RKHS classes without localization.* In infinite-dimensional kernel bandits, a global one-round DEC over an RKHS ball can be infinite or vacuous unless the comparison is localized to the finite active experiment actually visited by the algorithm; see the end of Section 5.1 for details. Known routes to matching DEC-style lower bounds introduce this localization explicitly or impose regularity conditions that enforce it implicitly through a critical-radius or sub-root fixed point (Foster et al., 2023). Indexed Bellman information complexity keeps this localization inside the posterior-reference process itself.
- *Double counting in multi-armed bandits.* The hard prior and the ghost entropy already contain the cost of identifying one arm among K . Therefore, after a Fano or posterior-reference calculation has produced the logarithmic entropy term, appending an independent DEC “estimation” penalty charges the same information budget a second time. This is not a statement that DEC-based algorithms are necessarily suboptimal for multi-armed bandits; it is a statement that the entropy appearing in the lower certificate and the estimation term appearing in a one-step upper relaxation should not be added as independent costs.
- *Collapsed lower bounds in linear bandits.* A one-round coefficient obtained by collapsing the adaptive trajectory to a single action distribution can leave a localization or logarithmic gap in the lower bound. The gap is not intrinsic to the linear-bandit experiment: it is introduced by replacing the posterior trajectory by a static action law. The indexed Bellman formulation keeps the posterior covariance, the information telescope, and the ghost quantile on the same reference path.
- *Representational ambiguity for Bellman rank.* Bellman rank is a representation-relative algebraic certificate: it depends on a hypothesis class, value-function class, roll-in distribution, discrepancy, and Bellman-error witness (Jiang et al., 2017). Without fixing these observable witnesses, bounded Bellman rank is not itself a DMSO model class and does not determine an intrinsic DEC. Once a bilinear-class witness and its estimation complexity are specified, DEC-style questions become meaningful (Du et al., 2021). Thus, the lower bound requires a strong specification and a substantial gap in model complexity.

Definition 4.4 (Tight indexed Bellman representation). *Fix a problem family, a horizon sequence, and a class of admissible indices χ . An indexed Bellman representation is tight if, for the relevant initial states s_1 and for some prior-index pair (μ, χ) , there exist a regret supersolution $\bar{V}$ and Bellman-Fano lower certificates $\bar{C}, \underline{\mathcal{E}}$ on the same state space such that, up to the stated universal constants or logarithmic factors,*

$$\bar{V}_1(s_1) \lesssim Tr_\star(\mu, \chi), \quad \mathfrak{R}_T^\star \gtrsim Tr_\star(\mu, \chi),$$

where $r_\star(\mu, \chi)$ is the critical average-regret radius in (48).

This definition is the representation-level replacement for asking whether a one-round DEC coefficient is universally tight. The question is not whether a single static coefficient is always exact, but whether the same Bellman-sufficient state supports both an upper supersolution and a posterior-reference quantile lower certificate at the same critical radius.

5 Applications and examples

5.1 Finite-action kernel bandits: GP-UCB, GP-E2D, and AMS/EBO

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS. What becomes finite is the marginal experiment observed through the active actions. This finite marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR belief over the optimal active action.

Problem and finite active marginal. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain $\mathcal{X}_{\text{all}}$, with kernel k satisfying $k(x, x) \leq \kappa^2$. The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function $f^\star \in \mathcal{H}_k$ satisfies $\|f^\star\|_{\mathcal{H}_k} \leq B$, and at round t the learner chooses $X_t \in \mathcal{X}$ and observes

$$Y_t = f^\star(X_t) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \quad (49)$$

where the noise is conditionally R -sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter $\lambda > 0$; the frequentist confidence event is obtained by the usual self-normalized argument.

Let $F = (f(x_1), \dots, f(x_n)) \in \mathbb{R}^n$ be the active reward vector and let $K_{\mathcal{X}}$ be the $n \times n$ kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \quad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history H_{t-1} , the algorithmic posterior on F is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For $x_i \in \mathcal{X}$, write $m_t(x_i) = e_i^\top m_t$ and $s_t^2(x_i) = e_i^\top \Sigma_t e_i$. If $X_t = x_i$, the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)} \{Y_t - m_t(x_i)\}, \quad (50)$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \quad (51)$$

This is a finite marginal posterior update. It should not be read as an assumption that $f^\star$ has n parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector F .

Model-index information and action-index information. For the model-index specialization, take the index to be the active reward vector $Y_{\text{MAIR}} = F$. The one-step GP/MAIR information at state H_{t-1} is

$$\mathcal{I}_t^{\text{GP}}(p) := \mathbb{E}_{x \sim p} \frac{1}{2} \log \left(1 + \frac{s_t^2(x)}{\lambda} \right). \quad (52)$$

Along a realized action sequence,

$$C_T^{\text{GP}} := \sum_{t=1}^T \mathcal{I}_t^{\text{GP}}(\delta_{X_t}) = \frac{1}{2} \log \det(I + \lambda^{-1} K_{X_{1:T}, X_{1:T}}), \quad (53)$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length- T designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^\star(F) \in \arg \max_{x \in \mathcal{X}} F_x, \quad q_t(a) := \mathbb{P}(A^\star = a \mid H_{t-1}).$$

Let ν_t^a be the exact conditional law of F given H_{t-1} and $A^\star = a$. It is generally a truncated Gaussian on the cone where a is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x, \lambda)(\cdot) \nu_t^a(dF),$$

for the observation at action x given the event $A^\star = a$. The marginal predictive is

$$P_{t,x} = \sum_{a \in \mathcal{X}} q_t(a) P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\text{AIR}}(p) := \mathbb{E}_{x \sim p} \sum_{a \in \mathcal{X}} q_t(a) D_{\text{KL}}(P_{t,a,x} \| P_{t,x}). \quad (54)$$

The conditional AIR regret is

$$\Delta_t^{\text{AIR}}(p) := \sum_{a \in \mathcal{X}} q_t(a) \mathbb{E}_{F \sim \nu_t^a} \mathbb{E}_{x \sim p} [F_a - F_x]. \quad (55)$$

The exact indexed chain rule gives

$$\sum_{t=1}^T \mathbb{E} \mathcal{I}_t^{\text{AIR}}(p_t) = I(A^\star; H_T) \leq H(A^\star) \leq \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

Algorithm 1: GP-UCB. GP-UCB uses the Gaussian marginal posterior (m_t, Σ_t) and chooses

$$X_t \in \arg \max_{x \in \mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \quad (56)$$

The following event is the frequentist calibration certificate:

$$\mathcal{E}_{\text{GP}} := \{|f^*(x) - m_t(x)| \leq \beta_t s_t(x) \text{ for all } t \leq T, x \in \mathcal{X}\}. \quad (57)$$

For finite $\mathcal{X}$, standard RKHS self-normalized concentration gives $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$ for the usual choice of β_t , for example

$$\beta_t \asymp B + R \sqrt{\log \det(I + \lambda^{-1} K_{\mathcal{X},t}) + \log(1/\delta)},$$

with the precise form depending on the normalization of k and the ridge parameter. The next theorem isolates the deterministic part of the GP-UCB proof.

Theorem 5.1 (GP-UCB realized-information bound). *On the event $\mathcal{E}_{\text{GP}}$, the GP-UCB rule (56) satisfies*

$$\text{Reg}_T(f^*) \leq 2\beta_T \sum_{t=1}^T s_t(X_t) \leq 2\beta_T \sqrt{T c_{\kappa,\lambda} C_T^{\text{GP}}}, \quad (58)$$

where

$$c_{\kappa,\lambda} := \frac{2\kappa^2}{\log(1 + \kappa^2/\lambda)}$$

and C_T^{GP} is the realized information gain in (53). Consequently, if rewards are bounded in $[0, 1]$ and $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$, then

$$\mathbb{E} \text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T c_{\kappa,\lambda} \mathbb{E} C_T^{\text{GP}}} + T\delta.$$

Proof. Let $x^* \in \arg \max_{x \in \mathcal{X}} f^*(x)$. On $\mathcal{E}_{\text{GP}}$, optimism gives

$$f^*(x^*) \leq m_t(x^*) + \beta_t s_t(x^*) \leq m_t(X_t) + \beta_t s_t(X_t).$$

Calibration at X_t gives $m_t(X_t) \leq f^*(X_t) + \beta_t s_t(X_t)$. Hence

$$f^*(x^*) - f^*(X_t) \leq 2\beta_t s_t(X_t) \leq 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T \sum_{t=1}^T s_t^2(X_t)}.$$

For $u \in [0, \kappa^2/\lambda]$, monotonicity of $u/\log(1+u)$ on a bounded interval gives

$$\lambda u \leq \frac{\kappa^2}{\log(1 + \kappa^2/\lambda)} \log(1 + u).$$

Applying this with $u = s_t^2(X_t)/\lambda$ and using (53) proves (58). The expectation bound adds the trivial regret bound T on $\mathcal{E}_{\text{GP}}^c$ and uses Jensen’s inequality. $\square$

GP–UCB as a MAIR decomposition. The theorem is exactly Lemma 3.9 with the model-index information surrogate $g_t = \frac{1}{2} \log(1 + s_t^2(X_t)/\lambda)$. The log determinant $\sum_t g_t = C_T^{\text{GP}}$ is the estimation-complexity telescope. The confidence multiplier β_t is the coefficient controlling the MAIR bracket. This is why GP–UCB is often practically strong: it follows the realized trajectory and pays the realized information gain, not the worst possible design information, even though its coefficient may be conservative in step-uniform minimax analyses.

Algorithm 2: GP–E2D. A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized one-round MAIR/DEC offset. Given a localized comparison set $\mathcal{F}_t$ containing plausible functions, define

$$\Delta_f(p) := \max_{x \in \mathcal{X}} f(x) - \mathbb{E}_{x \sim p} f(x),$$

and let $P_{f,x} = N(f(x), \lambda)$ while $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$ is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathcal{F}_t} \{ \Delta_f(p) - \gamma_t \mathbb{E}_{x \sim p} D_{\text{KL}}(P_{f,x} \| P_{t,x}) \}. \quad (59)$$

If $f^* \in \mathcal{F}_t$ and the optimized value in (59) is at most ε_t , then

$$\Delta_{f^*}(p_t) \leq \gamma_t \mathbb{E}_{x \sim p_t} D_{\text{KL}}(P_{f^*,x} \| P_{t,x}) + \varepsilon_t. \quad (60)$$

Summing (60) gives a regret bound once the fixed-truth KL terms are related to a posterior-ratio or information telescope. Under a well-specified Bayesian GP reference this relation holds in expectation by the MAIR chain rule. Under a fixed frequentist truth it requires calibration/localization. This is the uncertainty in GP–E2D’s role: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP log determinant is not automatic. An unlocalized offset over an infinite-dimensional RKHS ball can be infinite or vacuous; meaningful GP–E2D guarantees must localize the comparison set to the posterior-reference experiment.

Algorithm 3: AMS/EBO on the action index. AMS/EBO attacks the same bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is A^* . A robust AIR decision may be written schematically as

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{b \in \mathfrak{B}_t} \{ \Delta_b^{\text{AIR}}(p) - \gamma_t \mathcal{I}_b^{\text{AIR}}(p) \}, \quad (61)$$

where $\mathfrak{B}_t$ is a set of posterior/action-index beliefs around the current reference belief. If the chosen p_t satisfies

$$\Delta_t^{\text{AIR}}(p_t) \leq \gamma_t \mathcal{I}_t^{\text{AIR}}(p_t) + \varepsilon_t,$$

then the Bayes regret under the algorithmic reference is bounded by

$$\mathbb{E} \text{Reg}_T \leq \sum_{t=1}^T \gamma_t \mathbb{E} \mathcal{I}_t^{\text{AIR}}(p_t) + \sum_{t=1}^T \varepsilon_t \leq \gamma H(A^*) + \sum_{t=1}^T \varepsilon_t \quad (62)$$

when $\gamma_t \leq \gamma$. A ratio form gives the familiar Cauchy–Schwarz variant: if $(\Delta_t^{\text{AIR}}(p_t))^2 \leq \Psi_t \mathcal{I}_t^{\text{AIR}}(p_t)$, then

$$\mathbb{E} \text{Reg}_T \leq \sqrt{\left(\sum_{t=1}^T \mathbb{E} \Psi_t \right) H(A^*)}.$$

Since $H(A^*) \leq \log n$, this route scales with the finite decision index and the sequential AIR coefficient. It therefore has a natural hope of closing finite-action or low-dimensional action-set gaps that arise when GP-UCB's confidence multiplier is analyzed in the ambient RKHS. The route is different from GP-E2D: AMS/EBO uses the action-index posterior trajectory, whereas GP-E2D optimizes a one-round model-index offset whose relationship to the realized log determinant must be justified separately.

Position relative to the GP-UCB optimality debate. The original GP-UCB analysis is practical and trajectory-sensitive because it pays the realized information gain (53), or a worst-case relaxation only at the final step (Srinivas et al., 2010). Later work asks whether the remaining suboptimality comes from the algorithm or from the confidence analysis; in particular, Vakili et al. (2021b) ask for sharper online RKHS confidence intervals and sharper finite-action kernel-bandit guarantees. The indexed view separates the issue into three mechanisms. GP-UCB has the strongest realized-information geometry but may have a conservative MAIR coefficient. AMS/EBO can keep the action-index AIR telescope $I(A^*; H_T) \leq \log n$ and therefore has a principled route to finite-decision complexity, provided the robust bracket can be controlled. GP-E2D directly optimizes a one-round DEC coefficient, but an unlocalized DEC can be vacuous and a localized DEC must be matched to the same posterior-reference trajectory. Thus these algorithms should not be compared by a single universal coefficient; they are three relaxations of the same indexed Bellman identity with different estimation and coefficient mechanisms.

Why global RKHS DEC can be vacuous. Consider a Gaussian RKHS bandit with noise variance σ^2 , feature map $\phi(x) \in \mathcal{H}$ satisfying $\|\phi(x)\|_{\mathcal{H}} \leq 1$, and suppose the action set contains points $\{x_j : j \geq 1\}$ with $\phi(x_j) = e_j$, where $\{e_j\}_{j \geq 1}$ is orthonormal in $\mathcal{H}$. Let the reference model be $f_0 \equiv 0$, and let $\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq B\}$. For any one-round decision distribution p , define

$$q_j := \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}}^2.$$

Since

$$\sum_{j \geq 1} q_j \leq \mathbb{E}_{x \sim p} \|\phi(x)\|_{\mathcal{H}}^2 \leq 1,$$

there are directions with arbitrarily small q_j . For the alternative $f_j = B e_j$, the optimal action is x_j , while

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) = B - B \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}} \geq B(1 - \sqrt{q_j}),$$

and the one-round KL information against f_0 is

$$\mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \parallel N(0, \sigma^2)) = \frac{B^2}{2\sigma^2} q_j.$$

Hence, for every finite offset coefficient γ , one may choose j with q_j so small that

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) - \gamma \mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \parallel N(0, \sigma^2)) \geq cB$$

for a numerical constant $c > 0$. The constrained version has the same pathology: for any fixed information radius, a sufficiently rare direction has information inside the radius while retaining regret of order B . Thus an unlocalized one-round DEC over the entire RKHS ball need not reflect the effective difficulty of the finite experiment being run. The meaningful object is the finite active marginal together with its posterior-reference trajectory. Indexed Bellman information complexity keeps this localization inside the state process, rather than appending a separate DEC localization after the fact.

GP–E2D offset must be localized. The model-index offset in (59) is meaningful only relative to a comparison set and a reference law. If the comparison set is the entire infinite-dimensional RKHS ball, then a one-round adversary can place mass in directions receiving arbitrarily small action probability. The resulting regret remains order one while the one-step information is arbitrarily small, making the global offset vacuous. If the comparison set is instead the finite active posterior neighborhood, the offset may be finite and useful. The point is that the localization is part of the representation and posterior-reference trajectory; it should not be appended as an independent estimation term after a separate log-determinant analysis.

5.2 Multi-armed bandits lower bound via reference history

Consider $K \geq 3$ Gaussian arms with unit noise variance. For $j \in [K]$, environment $\omega = j$ has mean $\theta_j(a) = \Delta \mathbb{1}\{a = j\}$. The optimal arm is j , and one-step regret is

$$\ell_j(a) = \Delta \mathbb{1}\{a \neq j\}.$$

Let $Y = \Omega$ be uniformly distributed on $[K]$. Here Y is the action index, rather than the model index used in a direct one-step DEC application.

Lemma 5.2 (MAB ghost entropy). *For every algorithm and every reference history H'_T independent of $Y \sim \text{Unif}([K])$,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K}.$$

Proof. For a fixed reference action sequence $a'_1, \dots, a'_T$, let $N'_y = \sum_{t=1}^T \mathbb{1}\{a'_t = y\}$. The average regret against target y is $\Delta(1 - N'_y/T)$. The event $\bar{L}_y \leq \Delta/2$ implies $N'_y \geq T/2$. Since Y is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y \mid H'_T] = \frac{1}{K} \sum_{y=1}^K N'_y = \frac{T}{K}.$$

Markov's inequality gives $\mathbb{P}(N'_Y \geq T/2 \mid H'_T) \leq 2/K$. Averaging over H'_T proves the claim. $\square$

Lemma 5.3 (MAB information bound). *There is a universal constant $c_0 > 0$ such that if $\Delta^2 T/K \leq c_0$, then every adaptive algorithm satisfies*

$$I(Y; H_T) \leq \frac{1}{4} \log(K/2),$$

after adjusting constants for all $K \geq 3$.

Proof. Let $\mathbb{P}_0$ be the law under the zero-mean reference model, $\mathbb{P}_j$ the law under environment j , and $\bar{\mathbb{P}} = K^{-1} \sum_j \mathbb{P}_j$. Since $\bar{\mathbb{P}} \geq K^{-1} \mathbb{P}_j$, the likelihood ratio $d\mathbb{P}_j/d\bar{\mathbb{P}}$ is bounded by K . The inequality $D_{\text{KL}}(P\|Q) \leq (2 + \log K)H^2(P, Q)$ under $Q \geq K^{-1}P$, together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K} \sum_j D_{\text{KL}}(\mathbb{P}_j \|\bar{\mathbb{P}}) \leq 4(2 + \log K) \frac{1}{K} \sum_j H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using $H^2(P, Q) \leq D_{\text{KL}}(Q\|P)$ and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K} \sum_j H^2(\mathbb{P}_j, \mathbb{P}_0) \leq \frac{1}{K} \sum_j D_{\text{KL}}(\mathbb{P}_0 \|\mathbb{P}_j) = \frac{\Delta^2}{2K} \mathbb{E}_0 \sum_{t=1}^T \sum_{j=1}^K \mathbb{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Choosing c_0 small enough proves the result. $\square$

Theorem 5.4 (MAB lower bound). *For Gaussian K -armed bandits with unit noise variance,*

$$\mathfrak{R}_T^* \geq c\sqrt{KT}$$

for a universal constant $c > 0$ whenever $T \geq K \geq 3$.

Proof. Choose $\Delta = c_1\sqrt{K/T}$ with c_1 small enough for Lemma 5.3. Let $r = \Delta/2$. Lemma 5.2 gives $p_r \leq 2/K$, while Lemma 5.3 gives $T\tilde{C} = I(Y; H_T) \leq \frac{1}{4}\log(K/2)$. For c_1 small enough, condition (41) holds. Theorem 4.1 yields Bayes regret at least $Tr/2 = T\Delta/4 = c\sqrt{KT}$. Yao's principle gives the minimax lower bound. $\square$

On the upper-bound side, action-index AIR avoids the double counting that can occur in E2D. E2D may pay a K -factor both through model estimation and through the DEC term, whereas the action-index AIR identity in Theorem 3.5 incurs only a $\log K$ information-gain term and pays the K -factor through the AIR coefficient (Xu and Zeevi, 2025). This nearly matches the lower bound.

5.3 Linear bandits lower bound via reference history

Consider stochastic linear bandits on the Euclidean unit ball. The learner chooses $A_t \in \mathbb{R}^d$ with $\|A_t\|_2 \leq 1$ and observes

$$O_t = \langle \theta, A_t \rangle + \xi_t, \quad \xi_t \sim N(0, 1),$$

where the noises are independent over time. We prove the standard $\Omega(d\sqrt{T})$ minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior and an adaptive information bound. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown parameter. Chen et al. (2024, Section 3.3) prove the same lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches recover lower bounds for linear bandits that are tighter than those obtained from DEC alone.

Let

$$V \sim \text{Unif}(\{\pm 1\}^d), \quad \theta = \Delta V.$$

The corresponding optimal action is

$$a^*(V) = \frac{V}{\sqrt{d}},$$

and the one-step regret of action $a \in \mathbb{R}^d$, $\|a\|_2 \leq 1$, is

$$\ell_V(a) = \Delta\sqrt{d} - \Delta\langle V, a \rangle.$$

Assume $T \geq d^2$ and choose $\Delta \leq 1/\sqrt{d}$, so that $\|\theta\|_2 = \Delta\sqrt{d} \leq 1$.

Lemma 5.5 (Linear ghost entropy). *There is a universal constant $c_{\text{gh}} > 0$ such that every deterministic reference action sequence $a'_1, \dots, a'_T$, with $\|a'_t\|_2 \leq 1$, satisfies*

$$\mathbb{P}_{V \sim \text{Unif}(\{\pm 1\}^d)} \left(\frac{1}{T} \sum_{t=1}^T (\Delta\sqrt{d} - \Delta\langle V, a'_t \rangle) \leq \frac{\Delta\sqrt{d}}{4} \right) \leq \exp(-c_{\text{gh}}d).$$

Consequently, the same bound holds after averaging over any random reference history H_T' whose induced action sequence is independent of V .

Proof. Let

$$\bar{a} := \frac{1}{T} \sum_{t=1}^T a'_t.$$

Since $\|\bar{a}\|_2 \leq T^{-1} \sum_t \|a'_t\|_2 \leq 1$, the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a} \rangle \leq \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4}.$$

The random variable $\langle V, \bar{a} \rangle = \sum_{i=1}^d V_i \bar{a}_i$ is a centered Rademacher sum with variance proxy $\|\bar{a}\|_2^2 \leq 1$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4}\right) \leq \exp\left(-\frac{9d}{32}\right).$$

Thus the first claim holds, for example with $c_{\text{gh}} = 9/32$. If the reference action sequence is random but independent of V , then conditioning on the reference history and applying the deterministic bound gives the second claim. $\square$

Lemma 5.6 (Adaptive hypercube information capacity). *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(V; H_T) \leq \Delta^2 T,$$

where $H_T = (A_1, O_1, \dots, A_T, O_T)$.

Proof. It suffices to condition on the learner's internal randomness, which is independent of V , and prove the result for deterministic adaptive algorithms. The general randomized case follows by averaging.

For $v \in \{\pm 1\}^d$, let $\mathbb{P}_v$ denote the trajectory law when $\theta = \Delta v$. We use the chain rule over coordinates:

$$I(V; H_T) = \sum_{i=1}^d I(V_i; H_T \mid V_1, \dots, V_{i-1}).$$

Fix a coordinate i and a prefix $u = (v_1, \dots, v_{i-1})$. Let $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ be the trajectory laws after averaging uniformly over the remaining coordinates $V_{i+1}, \dots, V_d$, conditional respectively on $V_i = +1$ and $V_i = -1$. Then

$$I(V_i; H_T \mid V_{<i} = u) = D_{\text{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where D_{JS} denotes the Jensen–Shannon divergence with equal weights. For any two probability measures P, Q ,

$$D_{\text{JS}}(P, Q) \leq \frac{1}{4} \left(D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel P) \right).$$

By convexity of KL in its second argument and the log-sum inequality, the KL between the mixtures $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ is bounded by the average KL between coupled vertices that differ only in coordinate i . More explicitly, if $w \in \{\pm 1\}^{d-i}$ denotes the suffix and

$$v^+(u, w) = (u, +1, w), \quad v^-(u, w) = (u, -1, w),$$

then

$$D_{\text{KL}}(\mathbb{P}_{u,+} \parallel \mathbb{P}_{u,-}) \leq \mathbb{E}_w D_{\text{KL}}(\mathbb{P}_{v^+(u,w)} \parallel \mathbb{P}_{v^-(u,w)}),$$

and the analogous reverse inequality also holds.

For two fixed vertices v^+ and v^- that differ only in coordinate i , the adaptive KL chain rule gives

$$\begin{aligned} D_{\text{KL}}(\mathbb{P}_{v^+} \parallel \mathbb{P}_{v^-}) &= \frac{1}{2} \mathbb{E}_{v^+} \sum_{t=1}^T \left(\Delta \langle v^+ - v^-, A_t \rangle \right)^2 \\ &= 2\Delta^2 \mathbb{E}_{v^+} \sum_{t=1}^T A_{t,i}^2. \end{aligned}$$

Here the action kernel contributes no KL because the algorithm uses the same decision rule under both models; only the Gaussian observation kernel changes. Similarly,

$$D_{\text{KL}}(\mathbb{P}_{v^-} \parallel \mathbb{P}_{v^+}) = 2\Delta^2 \mathbb{E}_{v^-} \sum_{t=1}^T A_{t,i}^2.$$

Combining the preceding displays yields

$$I(V_i; H_T \mid V_{<i} = u) \leq \Delta^2 \mathbb{E} \left[\sum_{t=1}^T A_{t,i}^2 \mid V_{<i} = u \right],$$

where the expectation on the right is under the original hypercube prior and the algorithm's trajectory law. Averaging over $V_{<i}$ and summing over i gives

$$\begin{aligned} I(V; H_T) &\leq \Delta^2 \mathbb{E} \sum_{t=1}^T \sum_{i=1}^d A_{t,i}^2 \\ &= \Delta^2 \mathbb{E} \sum_{t=1}^T \|A_t\|_2^2 \\ &\leq \Delta^2 T. \end{aligned}$$

This proves the claim. $\square$

Remark 5.7 (Why the fixed-design log-determinant argument is not used). *For a Gaussian prior, posterior conjugacy gives the adaptive information identity*

$$I(\Theta; H_T) = \frac{1}{2} \mathbb{E} \log \det \left(I + \Sigma_0^{1/2} \left(\sum_{t=1}^T A_t A_t^\top \right) \Sigma_0^{1/2} \right),$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning on that realized design matrix controls only the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma 5.6 therefore uses a coordinatewise adaptive KL argument instead.

Theorem 5.8 (Linear bandit lower bound). *For stochastic linear bandits on the Euclidean unit ball with unit Gaussian noise,*

$$\mathfrak{R}_T^* \geq c d \sqrt{T}$$

for a universal constant $c > 0$, whenever $T \geq d^2$.

Proof. Assume first that d is larger than a sufficiently large universal constant. Choose

$$\Delta = c_1 \sqrt{\frac{d}{T}},$$

where $c_1 > 0$ is a sufficiently small universal constant. Since $T \geq d^2$, choosing $c_1 \leq 1$ ensures

$$\Delta \leq \frac{1}{\sqrt{d}},$$

so the prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta \sqrt{d}}{4}.$$

Lemma 5.5 gives the reference ghost-good probability bound

$$p_r \leq \exp(-c_{\text{gh}} d).$$

Lemma 5.6 gives the adaptive information bound

$$T\bar{C} = I(V; H_T) \leq \Delta^2 T = c_1^2 d.$$

Taking $c_1 > 0$ sufficiently small ensures

$$T\bar{C} + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r}.$$

The average-regret form of the quantile indexed information theorem, (41), therefore yields Bayes regret at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8} d\sqrt{T}.$$

Yao's principle then gives the same lower bound for the minimax regret.

For dimensions d below the universal threshold used above, the same conclusion follows, after reducing the universal constant, from the standard one-dimensional two-point lower bound applied to the first coordinate. Since d is then bounded by an absolute constant, $\Omega(\sqrt{T}) = \Omega(d\sqrt{T})$. This completes the proof for all d . $\square$

5.4 Bellman-rank as indexed representation

Low Bellman rank was introduced by Jiang et al. (2017) to show that contextual decision processes with suitable bilinear Bellman-error structure are PAC-learnable. In the present language, Bellman rank is a way to build a low-dimensional Bellman-sufficient representation. The following proposition is structural; it is not a new sharp lower bound for every Bellman-rank class.

Proposition 5.9 (Bellman rank gives an indexed representation). *Suppose a contextual decision process admits Bellman-error factorization of rank d in the sense of Jiang et al. (2017): for each hypothesis pair and policy, the relevant Bellman error can be written as an inner product of two d -dimensional feature vectors. If the observation model is dominated and the posterior predictive law is determined by these Bellman-error features together with the current posterior, then the process admits an indexed Bellman representation with state dimension controlled by the posterior over the d -dimensional features. Consequently, any information supersolution based on the induced linear-Gaussian or bounded-likelihood feature experiment yields a Bellman-Fano lower certificate through Theorem 4.3.*

Proof. The Bellman-error factorization supplies a finite-dimensional statistic sufficient to evaluate one-step Bellman errors for all candidate policies. If the predictive observation law and posterior update are functions of this statistic and the current posterior, then Definition 2.3 is satisfied with S_t equal to the posterior over these features and the accumulated planning state. The information and ghost-good recursions are then well-defined on the same state. Applying Theorem 4.3 gives the claimed certificate. The proposition is intentionally conditional because Bellman-rank learnability is a structural upper-bound statement; sharp lower bounds require a packing or ghost-entropy calculation inside the particular rank- d class. $\square$

This example clarifies the connection between the present framework and the broader perspective on representation learning developed in Li and Xu (2026); Xu (2026). The point is not that every low-Bellman-rank problem has the same minimax rate. The point is that a Bellman-rank factorization can provide the state on which the upper supersolution and the lower quantile-information certificate should be compared.

6 Conclusion

The paper argues that the natural unit of information complexity in interactive decision making is a Bellman representation equipped with an indexed posterior-reference process. Once the state and index are chosen, the same objects appear on both sides of the theory. Upper bounds are regret supersolutions or potential identities whose brackets are paid for by indexed information. Lower bounds are quantile-information certificates: the reference trajectory supplies both the information telescope and the ghost probability of small average regret. The critical radius is determined by

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r},$$

which is the interactive analogue of the classical local-prior-mass/Fano balance.

AIR and MAIR are two choices of index within the same log-gain Bellman supersolution. Choosing $\chi(\Omega) = A^*(\Omega)$ gives action-index AIR, which is often the right description of decision difficulty. Choosing $\chi(\Omega) = \Omega$, or the active reward vector in a bandit problem, gives model-index MAIR, which is often the convenient description of estimation geometry. UCB, E2D/DEC, and AMS/EBO then become three relaxations of the same indexed bracket: calibration and optimism, one-step offset optimization, and robust saddle control.

The kernel-bandit application illustrates why this separation matters. GP-UCB is practical because it keeps the realized log-determinant information gain of the actual trajectory. AMS/EBO, in an AIR form, can scale with the entropy of the finite or low-dimensional optimal-action index and the sequential AIR bracket. GP-E2D optimizes a one-round DEC offset, but its interaction with realized information gain is a theorem to be proved rather than a formal consequence of the offset definition; without localization a global RKHS DEC may be vacuous. Thus the right comparison is not a universal conversion coefficient. It is whether a single Bellman-sufficient state supports matching upper supersolutions and lower quantile certificates without charging estimation difficulty twice.

Several directions remain. The first is to identify representations for which the regret supersolution and the Bellman-Fano lower certificate match up to constants or logarithms. The second is to replace one-step DEC calculations by dynamically consistent Bellman relaxations when posterior evolution matters. The third is to sharpen the finite-action kernel theory by proving which of GP-UCB, AMS/EBO, or GP-E2D attains the finite-marginal information scale under the weakest

calibration assumptions. These are profound representation-level questions. The goal is not a universal one-shot notion of dimension, but a distribution-dependent dynamic theory of state spaces that accounts for estimation, decision, and lower-bound entropy costs once, rather than repeatedly.

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